Summary in Graph

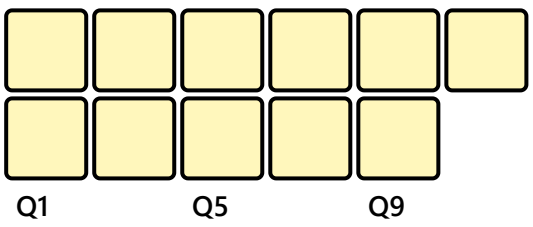
Exam Summary (GO Classes GATE CS | Data Structures | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	11 (2 + 9)
Total Marks:	20 (2 + 18)
Exam Duration:	35 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (22) out of 216 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

In-order traversing a tree resulted in E A C K F H D B G; the preorder traversal would return.

A. F A E K C D B H G

B. F A E K C D H G B

C. E A F K H D C B G

D. F E A K D C H B G

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Let $C(X)$ be the number of leaves in a binary tree rooted at T. Assume that $\text{IsLeaf}(T)$ returns 1 if T is a leaf. Which of the following observations leads to a recursive implementation?

A. $C(T) := C(T.\text{Left}) + C(T.\text{Right})$

B. $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + 1$

C. $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + \text{IsLeaf}(T)$

D. $C(T) := C(T.\text{Left}) + C(T.\text{Right}) + \text{IsLeaf}(T) + 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose that a binary min-heap stores six elements with priorities $10, 20, 30, 40, 50,$ and 60 in its array A. What is the largest of these items that could be stored in $A[1]$? (indexing starts from zero)

A. 20

B. 30

C. 40

D. 50

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose you wanted to insert the following elements into a binary search tree of integers: $90, 3, 16, 7, 12, 38, 50$. Which of the following insertion orders will result in a *complete binary tree?*

A. $16, 7, 3, 12, 50, 38, 90$

B. $16, 38, 3, 12, 50, 90, 7$

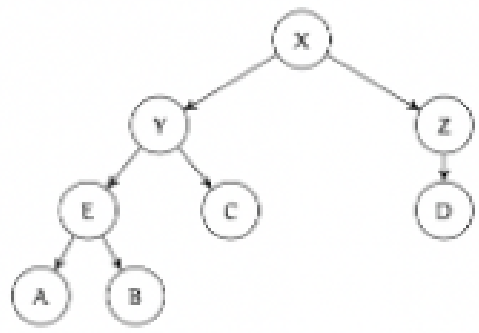
C. $7, 16, 3, 90, 12, 50, 38$

D. $16, 3, 12, 50, 90, 7, 38$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

In a level order traversal of a tree, nodes are traversed level-wise from the root and left to right in a level. For example, the level order traversal of the binary tree shown below is $\mathrm{X}, \mathrm{Y}, \mathrm{Z}, \mathrm{E}, \mathrm{C}, \mathrm{D}, \mathrm{A}, \mathrm{B}$.



Which one of the following combinations uniquely defines a binary tree?

A. Level-order and pre-order traversals

B. Level-order and in-order traversals

C. Level-order and post-order traversals

D. Pre-order and post-order traversals

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

We consider log to the base of 2. Thus for $\log \mathrm{N}$ to be a natural number, N must be a power of 2. Let N be an arbitrary number with $\mathrm{N} > 1$. Which one of the following statements is true? (Assume root is at height zero)

A. A binary tree of height $\log \mathrm{N}$ can have exactly N nodes.

B. A binary tree of height $\log \mathrm{N}$ can have exactly $\log \mathrm{N}$ nodes.

C. A binary tree of height $\log \mathrm{N}$ can have exactly $\mathrm{N} \log \mathrm{N}$ nodes.

D. A binary tree of height $\log \mathrm{N}$ can have exactly N^2 nodes.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following equation, where N is the number of nodes in a particular binary search tree, and $d\left(x_i\right)$ is the depth of node x_i in the tree. What does this equation compute?

Let All nodes in the tree are equally likely to be searched for.

$$\left(\sum_{i=1}^{\mathrm{N}} d\left(x_i\right)\right) / \mathrm{N}$$

A. The average depth of a node in a specific BST with N nodes

B. The average depth of a node in the average BST with N nodes

C. The total depth of all nodes in a specific BST with N nodes

D. The average total depth of all nodes in all BSTs with N nodes

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Suppose we have a complete binary tree, X, that is not a heap, and we wish to heapify it. The obvious thing to do is just insert all nodes of X into a new binary heap Y. This works, but it doubles the space requirement. Suppose we'd prefer to heapify in place. Which of the following procedures will convert X into a heap (containing all of the original values)?

A. Sink (heapify down) all nodes in level order (first the root, then its left child, then the right child of the root, etc).

B. Swim (heapify up) all nodes in level order.

C. Swim all nodes in reverse level order.

D. Sink all nodes in reverse level order.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following is/are true?

A. Suppose T is a completely filled binary search tree with 7 nodes, the worst-case number of comparisons for a successful find operation in T is 3

B. Suppose T is a completely filled binary search tree with 7 nodes. the best case number of comparisons for a successful find operation in T is 1

C. Suppose T is a binary search tree with 7 nodes, only one of which is a leaf. suppose all keys are equally likely, the integer closest to the average case number of comparisons for a successful find operation in T is 4

D. Suppose T is a completely filled binary search tree with 7 nodes. suppose all keys are equally likely. The integer closest to the average case number of comparisons for a successful find operation in T is 2

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Number of AVL tree(s) containing elements with keys $2, 4, 6, 8,$ and 10.

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following is is/are true ?

A. The depths of any two leaves in a max heap differ by at most 1.

B. Inserting into an AVL tree with n nodes requires $\Theta(\log n)$ rotations.

C. The height of any binary search tree with n nodes is $O(\log n)$.

D. We can convert a heap to Binary Search Tree in $\Theta(n)$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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